

Changes in Elbow Stress and Ball Velocity During Reduced Effort Pitching

A Marker-Based Motion Capture Analysis

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Background: Baseball pitchers often participate in throwing programs that involve throwing at reduced effort levels to gradually increase the amount of stress experienced across the elbow. It is currently unknown how reduced effort pitching compares with maximum effort with respect to elbow stress and ball velocity.

Purpose/Hypothesis: The purpose was to determine the correlation between elbow stress and ball velocity with reduced effort pitching. We hypothesized that decreased perceived effort would disproportionately correlate with elbow stress and ball velocity.

Study Design: Descriptive laboratory study.

Methods: Ten healthy male high school baseball pitchers threw 5 pitches from a regulation pitching mound at 3 effort levels: maximum effort, 75% effort, and 50% effort. Elbow stress, specifically elbow varus torque, was calculated for all pitches using a validated marker-based 3-dimensional motion capture system. Ball velocity was measured using a Doppler radar gun. Intrathrower variability was calculated for each effort level.

Results: Elbow stress and ball velocity decreased with reduced effort throws ($P < .001$ and $P = .003$, respectively). However, the reductions in elbow stress and ball velocity were not proportional. At 75% effort throws, elbow stress measured 81% (intraclass correlation coefficient [ICC], 0.95), and ball velocity measured 90% (ICC, 0.80) of maximum, respectively. At 50% effort throws, elbow stress measured 75% (ICC, 0.93), and ball velocity measured 85% (ICC, 0.87) of maximum. Intrathrower reliability was excellent for elbow stress and ball velocity, with all ICCs ≥ 0.80 .

Conclusion: Pitching at a reduced effort level resulted in decreased elbow stress and ball velocity. However, for every 25% reduction in perceived effort, elbow stress decreased by a mean 13%, and ball velocity decreased 7.5%. When baseball pitchers attempt to throw at a reduced effort of maximum, throwing metrics do not decrease proportionately.

Clinical Relevance: While pitching at a reduced effort of maximum decreases elbow stress and ball velocity, the decrease is not proportional, subjecting the elbow to more stress than intended. This has significant clinical importance to pitchers, coaches, and medical professionals in the setting of injury prevention and return to sports.

Keywords: reduced effort; pitching; elbow varus torque; elbow stress; baseball; intrathrower reliability

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Multiple studies have demonstrated that baseball injuries are increasingly prevalent at all levels of play, including youth, collegiate, and professional athletes.^{3,8,14,21} While these injuries can occur to any part of the body, upper extremity injuries in pitchers continue to be one of the most common reasons for time on the injured list, excluding participation in practice and games.^{1,7,19,25,26,28} In a study of nearly 50,000 players in professional baseball, 39% of all injuries occurred in pitchers who represented 53.3% of time a player or players were listed on an active professional baseball roster. Additionally, 39% of all injuries involved the upper extremity.³ Specifically, elbow injuries were particularly devastating to the throwing athlete, ending the athlete's season at a high rate and often requiring surgical treatment. Injury to the ulnar collateral ligament resulted in a season-ending rate of 60.0%.

Furthermore, elbow sprains, elbow medial epicondylitis, ulnar neuritis, and forearm flexor pronator strain diagnoses resulted in season-ending rates of 38.9%, 21.3%, 20.7%, and 20.0%, respectively.³ The injury itself is likely due to the repetitive high valgus forces experienced on the varus-stabilizing structures of the medial elbow, most notably the medial ulnar collateral ligament, during the late cocking and early acceleration phases of throwing.^{5,13} Other factors contributing to the rising trend of elbow injuries include higher workload/throwing volume,¹² increasing throwing velocity,^{18,30} and improper throwing mechanics.^{5,11} While much work has been done in an effort to reduce injury rates, these injuries continue to occur with increasing frequency.

To properly prepare these athletes for the high demands of pitching, a great deal of work has gone into developing interval throwing programs that gradually raise the amount of stress that the elbow sees over time as a lead-up to competition.^{18,23,24,31,33} These throwing programs are commonly used to prepare healthy athletes for the season ahead and for the injured athlete working on returning to play after injury. Although these programs vary to some extent, the general principles of gradually increasing throwing effort, distance, and volume over time are consistent across programs. While much work has been done on the effect of throwing distance on elbow stress,^{14,15,18} less is known about the effect of the player's effort. During the preliminary stages of these programs, pitchers are often instructed to throw at a reduced effort of maximum (ie, "50% effort" or "75% effort") with the goal of lowering the stress on the athlete's elbow.²³ This is typically done for the flat-ground portion of the program as well as the mound-throwing portion. Using a radar gun to analyze partial-effort pitching programs, Lizzio *et al*²⁰ demonstrated that pitchers throw at significantly higher velocity than intended. Slenker²⁹ showed that partial-effort throwing decreases loads experienced by the shoulder and elbow, while Fleisig *et al*¹⁶ demonstrated lowered velocity, bodily forces, and elbow torque in collegiate pitchers as they threw at efforts less than maximum. In a recent study, Melugin *et al*²³ evaluated reduced effort throwing metrics during a flat-ground long-toss program with a wearable sensor applied to the throwing arm. Ultimately, they concluded that for every 25% reduction in effort, elbow torque decreased only 7%.

While this work was informative for long-toss and flat-ground throwing, less is known about how decreased effort affects elbow torque when pitching from the pitcher's mound. Because mound throwing is a critical portion of the throwing progression for pitchers and typically includes a reduced effort component, a better understanding of the relationship of elbow torque and effort in this setting is needed. Accordingly, the primary goal of this work was to explore the relationship of elbow torque and pitch effort using the gold standard of marker-based motion capture. Specifically, we sought to (1) determine whether throwing metrics (specifically elbow varus torque and ball velocity) correlate with reduced effort pitching, (2) ascertain whether metrics decrease proportionally while pitching at a reduced effort, and (3) evaluate intrathrower

variability when throwing at a reduced effort. We hypothesized that decreased perceived effort would disproportionately correlate with throwing metrics (elbow stress and ball velocity) and that intrathrower reliability would be acceptable.

METHODS

Participants

After approval from our institutional review board, 10 healthy right-handed male baseball pitchers volunteered to participate. Eligibility criteria for the study were defined as follows: age 15 to 18 years, currently active in competitive baseball as a pitcher at a high school level, no current musculoskeletal concerns or functional limitations, full range of motion of bilateral upper and lower limbs, and willingness to participate in the study. Informed consent was obtained from each participant and parent or legal guardian before data collection.

Testing Procedure

Participants were asked to wear compression shorts and no shirt, if comfortable, or to wear a provided tank top and shorts. Each was given as much time as needed to warm up before throwing off a pitching mound. As fully described previously,⁴ participants were outfitted with retroreflective markers as follows (Figure 1): on the throwing arm (index metacarpal, third metacarpal, fifth metacarpal, radial and ulnar styloids, medial and lateral epicondyles, acromioclavicular joint), on the nonthrowing arm (acromioclavicular joint, lateral epicondyle, carpals), centrally (C7 and T10 spinous processes, manubrium, xiphoid process), and bilaterally (calcaneus, lateral malleolus, midfoot between the second and third metatarsals, tibia, lateral femoral epicondyle, femur, anterior and posterior iliac spine, sacrum). Static data collection was obtained with the pitcher positioned with his forearms in an anatomic neutral position with additional markers placed bilaterally on the medial femoral condyles and medial malleoli. The joint centers of the knee and ankle were defined as the midpoint of the femoral condyles and malleoli, respectively.

Each pitcher threw fastballs at maximum effort until 5 strikes were recorded. A strike was defined as any pitch thrown over home plate through a simulated strike zone into a net. For consistency, participants were asked to throw only fastballs. After maximum effort fastballs were recorded, participants were asked to throw 5 additional fastballs at 75% and 50% of their maximum effort, respectively. For these pitches, a strike was encouraged but not necessary. To simulate real-world throwing programs, this reduction in perceived effort was based on each pitcher's subjective assessment of his maximum effort. No additional coaching or instruction was provided. All pitches were thrown from a standard elevation pitching mound to a strike zone target located at a regulation distance of 60 ft 6 in (18.4 m) using a regulation size baseball of 5 oz

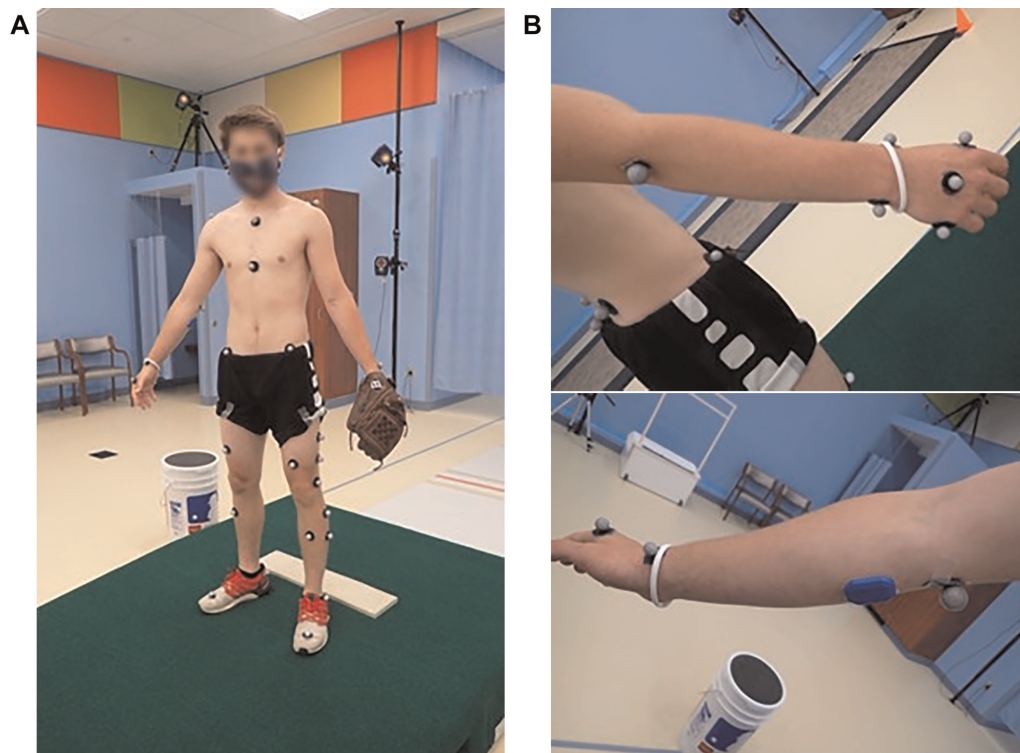


Figure 1. Position of retroreflective markers on (A) the participant and (B) his throwing arm.

(142 g). For each pitch, ball velocity was measured using a Doppler radar gun (JUGS Sports). The 3-dimensional marker trajectories of the reflective markers were collected at 500 Hz using a 14-camera motion-capture system (Raptor 12HS; Motion Analysis Corp) and a commercial software program (Cortex Version 8.1; Motion Analysis Corp).

Joint kinematics and kinetics of the throwing arm were calculated using a custom software program implemented in Visual 3D (C-Motion Inc). Position data were filtered using a 20-Hz fourth-order Butterworth low-pass filter. Segment coordinate systems and rotation sequences were defined in accordance with recommendations from the International Society of Biomechanics.³⁴ Joint kinetics were derived via inverse dynamics.¹⁷ The inertial properties¹⁰ of the hand, forearm, and arm were input into the model. The force of the baseball was modeled as a point mass of 142 g (5 oz) at the third metacarpal marker until the ball was released. After ball release, the force from the ball mass was omitted from the model. All metrics were resolved to the model-generated segment coordinate systems utilizing the upper arm, forearm, and hand segments with respect to one another. All metrics were normalized to each pitcher's height and weight, producing torque units labeled N·m/N·m.

The elbow varus torque (elbow stress) and ball velocity of the 75% and 50% perceived effort throws were reported as a percentage of the maximum effort throws. A 1-factor repeated measures analysis of variance and Fisher least significant difference test were performed to assess the comparison of ball velocity and elbow stress across the

different exertion levels, respectively. The mean values across the pitches for each pitching athlete were utilized for all statistics to preserve the degrees of freedom. All statistical testing was performed using SAS Version 9.4 (SAS Institute). Intrathrower reliability, or the agreement among pitchers, was calculated for elbow stress and ball velocity at each perceived effort level. Each pitch thrown by participants at each perceived effort level was compared with the others using mean error as a percentage with 95% CIs. Additionally, reliability measures were reported via intraclass correlation coefficients (ICCs). Reliable agreement was classified as poor (<0.40), fair (0.40-0.59), good (0.60-0.74), or excellent (≥ 0.75) as previously described.^{18,23} Last, the coefficient of repeatability (COR) was reported for elbow stress and ball velocity at all effort levels. The COR is a measure of reliability representing the magnitude of expected differences among observations within an individual. As the COR approaches zero, there is a greater reliability of the measured metric.¹⁸

RESULTS

Ten healthy pitching athletes participated in the study. Their mean $\pm$ SD age was 16.7 ± 1.0 years; mean height, 1.9 ± 1.0 m; and mean body mass index, 22.4 ± 2.4 .

Normalized values are presented in Table 1. Elbow stress and ball velocity decreased with reductions in perceived effort ($P < .001$ and $P = .003$, respectively). Post

TABLE 1
Maximum and Perceived Reduced Efforts in Ball Velocity and Elbow Stress

Perceived Effort	Measured Elbow Stress		Ball Velocity	
	Mean $\pm$ SD, N·m	Maximum Effort: Mean $\pm$ SD, %	Mean $\pm$ SD, mph	Maximum Effort: Mean $\pm$ SD, %
100%	68.2 $\pm$ 20.1	100	73 $\pm$ 5	100
75%	55.3 $\pm$ 18.2	81 $\pm$ 7	67 $\pm$ 6	90 $\pm$ 6
50%	50.9 $\pm$ 17.9	74 $\pm$ 10	62 $\pm$ 8	85 $\pm$ 9
P value	<.001	<.05	.003	<.05

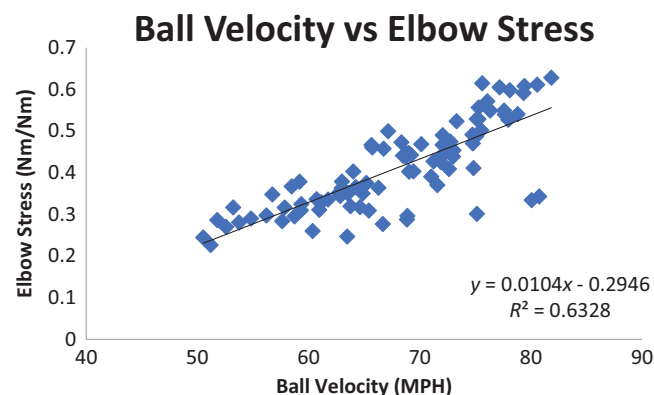


Figure 2. Elbow stress as a function of ball velocity for all pitches thrown. There was a significant increase in elbow stress as ball velocity increased; however, there was not a strong relationship with a linear model ($R^2 = 0.6328$).

TABLE 2
Intraclass Correlation Coefficients

Metric	Intraclass Correlation Coefficient
Elbow stress	
Maximum effort	0.94
75% effort	0.95
50% effort	0.93
Ball velocity	
Maximum effort	0.82
75% effort	0.80
50% effort	0.87

hoc analyses of elbow stress and ball velocity demonstrated that 75% and 50% perceived exertion levels yielded throwing metrics that were less than the maximum effort level ($P < .05$). For every increase in 1 mph of ball velocity, there was an observed increase of 0.0104 N·m/N·m in elbow stress ($R^2 = 0.633$) (Figure 2).

While elbow stress and ball velocity decreased with reduced effort throws, these metrics did not decrease proportionately (Table 1). For every 25% reduction in perceived effort, elbow stress decreased by a mean 13%, and ball velocity decreased by a mean 7.5%. The mean elbow stress measured 81% and 74% of maximum while reducing perceived effort to 75% and 50% of maximum, respectively.

Additionally, the mean ball velocity measured 90% and 85% of maximum while reducing perceived effort to 75% and 50% of maximum, respectively.

The intrathrower reliability was considered excellent, as the mean elbow stress error from all pitchers was $3.6\% \pm 1.5\%$ (95% CI, 2.6%-4.6%) and the mean ball velocity error was $3.0\% \pm 1.3\%$ (95% CI, 2.5%-3.5%). The ICCs were ≥ 0.80 for ball velocity and elbow stress at all levels of perceived effort (Table 2). Mean COR values for elbow stress were 8.02, 6.46, and 8.92 for maximum effort, 75% effort, and 50% effort, respectively. Mean COR values for ball velocity were 2.58, 3.29, and 5.59 at maximum effort, 75% effort, and 50% effort. Intrathrower elbow stress reliability was poorest at perceived 50% effort (COR, 8.92) (Figure 3). Similarly, intrathrower ball velocity reliability was poorest at perceived 50% effort (COR, 5.69) (Figure 4).

DISCUSSION

In this study analyzing metrics using 3-dimensional motion capture in pitchers throwing at reduced effort levels, as compared with maximum effort pitching, elbow stress and ball velocity decreased at 75% and 50% effort levels, but this decrease was not proportional. Intrathrower reliability was excellent, with ICC ≥ 0.93 for elbow stress and ≥ 0.80 for ball velocity at all effort levels. Ultimately, we accepted the hypothesis that reduced effort pitching would lower elbow stress and ball velocity but this reduction would not be proportional. Additionally, we accepted the hypothesis that intrathrower reliability would be acceptable.

The findings in the current study specific to pitchers throwing from a mound corroborate previous studies of throwing athletes, which have shown that reduced effort throwing results in decreased elbow stress and ball velocity.^{23,29,33} In a study evaluating metrics with reduced effort throwing during a long-toss program, Melugin et al²³ demonstrated a reduction in elbow varus torque with reduced effort throws. At maximum effort, the mean elbow varus torque was 72.7 N·m, which decreased to 67.3 N·m and 62.2 N·m at 75% and 50% of maximum efforts, respectively. Additionally, ball velocity decreased from 77.8 mph at maximum effort to 66.5 and 60.3 mph at perceived efforts of 75% and 50%, respectively. Fleisig et al¹⁶ also determined that perceived overhead throwing efforts of 75% and 50% resulted in consistently reduced ball velocity.

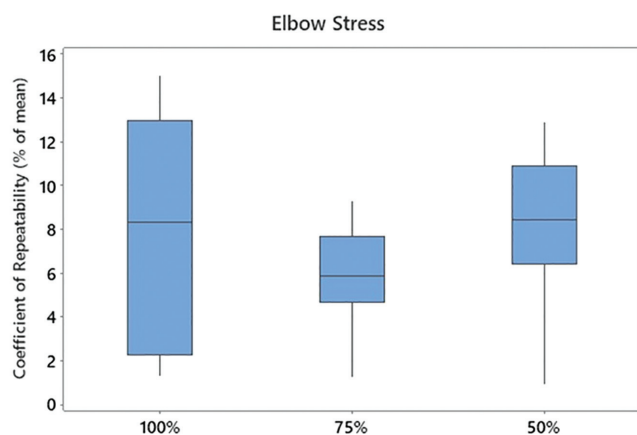


Figure 3. Elbow stress coefficient of repeatability at each effort level. Data are presented as mean $\pm$ SD (95% CI).

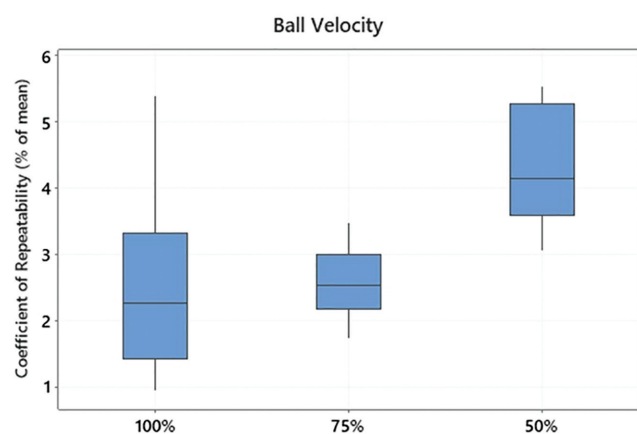


Figure 4. Ball velocity coefficient of repeatability at each effort level. Data are presented as mean $\pm$ SD (95% CI).

TABLE 3
Ball Velocity as Percentage of Maximum

Study	Ball Velocity as Percentage of Maximum	
	75% Effort	50% Effort
Current study: motion capture	90	85
Melugin et al ²³ : wearable sensor	86	78
Slenker et al ²⁹ : motion capture	90 (80% effort)	84 (60% effort)
Lizzio et al ²⁰ : radar gun	88.5	79.4
Fleisig et al ¹⁶ : motion capture	90	85

In a study describing a multiple-phase return-to-throwing program, Wilk et al³³ emphasized the importance of

TABLE 4
Elbow Stress/Torque as Percentage of Maximum

	Elbow Stress/Torque as Percentage of Maximum	
	75% Effort	50% Effort
Current study: motion capture	81	75
Melugin et al ²³ : wearable sensor	93	87
Slenker et al ²⁹ : motion capture	90 (80% effort)	75 (60% effort)
Lizzio et al ²⁰ : radar gun	88.6	76.3
Fleisig et al ¹⁶ : motion capture	85	75

monitoring throwing intensity, as athletes may have difficulty controlling throwing metrics at lower throwing intensities. In the current study, for every 25% reduction in perceived effort, mean elbow stress decreased only 13% and mean ball velocity decreased just 7.5%. These results are consistent with previous studies also demonstrating disproportionate decreases in throwing metrics with reduced effort levels^{14,23,29,33} (Tables 3 and 4). Melugin et al²³ showed, during a long-toss program, that for every 25% reduction in perceived effort, elbow varus torque decreased only 7% and ball velocity decreased by just 11%. In a study of pitchers throwing from a regulation baseball mound, Slenker et al²⁹ evaluated the relationship of valgus stress on the elbow and pitch velocity at various levels of effort. At 80% effort of maximum, elbow valgus load was 90%, and velocity was 90% of maximum. At 60% effort, elbow valgus load was 75%, and velocity was 84% of maximum.²⁹ Additionally, Fleisig et al¹⁶ demonstrated that perceived overhead throwing efforts of 75% and 50% resulted in ball velocity measuring 90% and 85% of maximum, respectively. The consistency in these data suggests that athletes may throw at a greater intensity than intended, subjecting the elbow to unintentional levels of stress.

The relationship of elbow stress and ball velocity has been controversial in the current literature, often owing to the different intrinsic characteristics of pitchers and associated biomechanics of delivery. Multiple studies have reported that higher pitch velocities place professional baseball players at risk for ulnar collateral ligament injury and reconstruction.^{2,6,9,27,32} Slowik et al³⁰ identified a positive correlation between higher ball velocity and higher elbow varus torque but indicated that pitcher-specific characteristics limit generalizability. Manzi et al²² found higher pitch velocity among professional pitchers a weak predictor of shoulder and elbow forces. However, when controlling for an individual pitcher, they found ball velocity was strongly predictive of forces across the shoulder and elbow, suggesting that pitchers with faster ball velocity do not necessarily experience greater elbow stress than do lower-velocity pitchers. The current

study adds context to these inconsistent reports on the nature of the relationship of elbow stress and ball velocity. In this work, both decreased with lower levels of effort, but they did not do so at the same rate. Accordingly, elbow stress does correlate with velocity to a certain degree, but this relationship is not perfectly proportional.

Intrathrower reliability has been evaluated in previous studies. Leafblad et al¹⁸ and Fleisig et al¹⁴ demonstrated increasing variability in throwing metrics as distances increased. Melugin et al²³ cited excellent intrathrower reliability with respect to elbow stress and ball velocity during a structured long-toss program at a set distance of 120 ft (ICC >0.75). The current study also revealed excellent intrathrower reliability with all throws being completed from a regulation pitching mound (ICC ≥0.80). The constant throwing distance in the current study likely contributed to the excellent intrathrower reliability. Interestingly, the increased variability at 50% perceived effort may be attributed to large decreases in partial velocity impairing the pitcher's ability to throw with proper technique, as suggested by Lizzio et al.²⁰ However, the overall high rates of reliability suggest that throwers are relatively consistent in their kinetics within each throwing scenario. As we continue to progress toward increased monitoring of throwing programs, it may be that not every single throw needs to be measured directly; that is, the metrics from a small sample of throws from a particular scenario can likely serve as a reliable surrogate for all of a pitcher's throws within that scenario. This information can be quite helpful in designing individualized throwing programs for our athletes, as it simplifies the process and logistics.

There are limitations to this study that merit discussion. First, athletes pitched in a motion analysis laboratory in a setting that was not game-like; it is also difficult to match the level of intensity of live competition, which together may have altered pitching efforts. Second, there was no formal instruction provided to pitchers with respect to their methods of warm-up. Enrolled participants were a small number of athletes limited to high school males; therefore, the results may not be applicable to more elite throwers at the collegiate or professional levels. All pitches occurred from a regulation mound, so the results may not be applicable to those pitching from a flat surface. Additionally, pitching efforts were performed in order of decreasing efforts, rather than randomizing each pitch to a specific effort. Last, the current study participants were composed of healthy, noninjured pitchers who may differ from pitchers recovering from an injury.

CONCLUSION

Pitchers were reliably able to reduce elbow stress and ball velocity with reduced effort throws. However, these metrics did not decrease proportionately. For every 25% reduction in perceived effort, mean elbow stress decreased only 13%, and ball velocity decreased just 7.5%. For all throwing scenarios, intrathrower reliability was considered excellent. These findings have clinical importance suggesting that pitchers may subject the elbow to more stress than

intended, and they must be considered by the athlete, coaches, and medical professionals in the setting of injury prevention and return to sports.

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